A Distributed SOS Program For Local Stability Analysis of Polynomial PDEs in the PIE Representation

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***Abstract*— It has recently been shown that the evolution of a state, described by a Partial Differential Equation (PDE), can be more conveniently represented as the evolution of the state's highest spatial derivative (the "fundamental state"), which lies in L_2 and has no boundary conditions (BCs) or continuity constraints. For linear PDEs, this yields a Partial Integral Equation (PIE) parametrized by Partial Integral (PI) operators mapping the fundamental state to the PDE state. In this paper, we show that for polynomial PDEs, the dynamics of the fundamental state can instead be compactly expressed as a distributed polynomial in the fundamental state, parametrized by a new tensor algebra of PI operators acting on the tensor product of the fundamental state. We further define a SOS parametrization of the distributed polynomial and use this to construct a distributed SOS program, for testing local stability of polynomial PDEs.**

I. Introduction

In this paper, we address the problem of representation and local stability analysis of polynomial PDEs. Such PDEs are commonly used to model physical processes including quantum mechanics, condensed matter, fluid dynamics, population growth, and wave propagation [1]. However, some features of the standard PDE representation create challenges for analysis and simulation. For example, consider Burgers Equation with the additional reaction term and Dirichlet BCs.

$$\begin{aligned} u_t(t,s) &= u_{ss}(t,s) - u(t,s)u_s(t,s) + ru(t,s) \\ u(t,0) &= u(t,1) = 0 \quad t \geq 0 \;\; s \in [0,1] \end{aligned} \tag{1}$$

Using integration by parts, the Poincaré inequality and invoking the BCs, the LF $V(u) = \|u\|_{L_2}^2 \geq 0$ can verify global stability of the zero solution if $r < \pi^2$ [2]. This approach works well for (1) since $\dot{V}(u)$ only depends on u, u_s, allowing the Poincaré inequality to be applied. More generally, this is not applicable and the representation of PDEs as polynomial functions of $u, u_s, \ldots, u_{s_n}$ complicates the stability analysis, even for simple LF candidates.

Now suppose a Dirichlet BC is replaced by a Neumann BC, $u_s(t,1) = 0$. Despite the PDE and LF candidate remaining the same, the derivative of this candidate along the solutions to (1) is different, meaning the previously obtained stability test is no longer valid. Furthermore, it is unclear from this representation how the BCs impact stability properties and how they can be accounted for when testing the fitness of a LF candidate.

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Because of these difficulties, most prior work focuses on the stability analysis of specific nonlinear PDEs and BCs. For example, stability conditions have been derived for the Navier-Stokes equations [3], [4], [5], [6] and the Kuramoto-Sivanshinsky equation with periodic BCs [7]. These results offer some, but limited insight into how to test stability of other nonlinear PDEs. Other work has studied limited classes of PDEs including the wave equation with a bounded nonlinear term [8] and a class of 2nd order parabolic PDEs [9], [10], [11]. These results focus on specific applications, and the proposed stability conditions may be challenging to enforce in practice.

To address the challenges posed by the representation of nonlinear PDEs, a new approach represents the dynamics of the PDE state as a PIE [12]. This representation expresses the dynamics of the PDE state only in terms of its highest spatial derivative, known as the "fundamental state". Unlike the PDE state, which lies in a Sobolev space, the fundamental state lies in L_2 and is free of BCs and continuity constraints. For linear PDEs, the PIE is parametrized by PI operators which map the fundamental state to the PDE state and it's derivatives [13], [14]. Through these operators, the BCs and continuity constraints are embedded in the PIE, rather than constraining the PDE state. More recently, this framework was extended to represent quadratic PDEs, where the PIE is parametrized by "Tensor-PI" operators, which act on tensor products of the fundamental state [15]. Another recent extension, enables the representation to also handle periodic BCs [16]. Leveraging the PIE representation, a suite of SDP tools have been developed for the stability analysis [13], [14], [17], estimation [18], and control [19], [20] of PDEs. These methods are implemented in the open-source software package, PIETOOLS [21].

Contribution: Inspired by the Sum-of-Squares (SOS) framework for polynomial ODEs [22], this article defines a PI representation of distributed polynomials in the fundamental state. This representation enables parametrization of the PIE, such that it can represent 1D polynomial PDEs, and define distributed semi-algebraic sets, like the L_2-ball. A SOS parametrization of the distributed polynomial is also defined, enabling positivity of distributed polynomials to be enforced through equality constraints. Leveraging such distributed polynomials, a distributed SOS program is constructed to test local stability on the L_2-ball. This test is suitable for all 1D polynomial PDEs with linear BCs.

II. Notation

For $\Omega := [a,b] \subset \mathbb{R}$, let $L_2^n[\Omega]$ and $L_\infty^n[\Omega]$ denote the spaces of square-integrable and bounded $\mathbb{R}^n$-valued func-

tions on Ω. For $k \in \mathbb{N}$, define the Sobolev space

$$H_2^k[\Omega] := \left\{x \in L_2[\Omega] : x_{s_\alpha} \in L_2[\Omega],\ \forall\, \alpha \le k\right\}$$

where the derivatives are denoted by $x_{s_\alpha} := \frac{\partial^\alpha}{\partial s^\alpha} x$, and $\mathcal{D}^k x(s) := [x(s)\ x_s(s)\ \cdots\ x_{s_k}(s)]'$. For $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$, let $u \otimes v \in \mathbb{R}^{mn}$ denote the Kronecker product. For $x \in L_2^m[a,b]$, $y \in L_2^n[c,d]$, define the tensor product $x \otimes y \in L_2^{mn}[[a,b]\times[c,d]]$ by $[x \otimes y](s_1,s_2) = x(s_1) \otimes y(s_2)$. For $d \in \mathbb{N}$ and $x \in L_2^n[\Omega]$, let $x^{\otimes d} := x \otimes \cdots \otimes x \in L_2^{n^d}[\Omega^d]$, and define the degree-$d$ distributed monomial basis $\mathcal{Z}_d(x) := [1\ x\ x^{\otimes 2}\ \cdots\ x^{\otimes d}]'$, omitting the degree-0 term when convenient. Let $\mathbb{R}[\boldsymbol{\theta}_k]$ denote the polynomial ring in k variables. For $G \in \mathbb{R}^{n\times m}$, row i is $G_{i,:}$. Let $\mathcal{S}_w := \{(i,j) \in \mathbb{N}^2 : i + j = w\}$ and $A_n \subseteq \{1,\dots,n\}^n$ denote a subset of all $n!$ permutations. For $\alpha \in A_n$ and $k,j \in \{1,\dots,n\}$, the indicator functions used are defined below, where $\mathbb{I}_1(\theta_1) := 1$ if $n = 1$.

$$\mathbf{1}(x) := \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{else} \end{cases} \qquad \mathbf{I}_\epsilon(x) := \begin{cases} 1 & \text{if } x \ge \epsilon \\ 0 & \text{else} \end{cases}$$

$$\gamma(\alpha_k, j) := \begin{cases} 1 & \text{if } j < k \\ 2 & \text{if } j \ge k \end{cases} \qquad \mathbb{I}_\alpha(\boldsymbol{\theta}_n) := \prod_{j=1}^{n-1} \mathbf{I}_0(\theta_{\alpha_{j+1}} - \theta_{\alpha_j})$$

Finally, nested integrals are compactly denoted by

$$\int_{\Omega^k} K(\boldsymbol{\theta}_k)d\boldsymbol{\theta}_k = \int_a^b \cdots \int_a^b K(\theta_1,\dots,\theta_k)d\theta_k \dots d\theta_1$$

III. A Map From Fundamental State to PDE State

For a domain $\Omega := [a,b]$, we consider the nth order, 1D, polynomial PDE

$$u_t(t,s) = c(s)^\top \mathcal{U}(u, u_s, \dots, u_{s_n}) \quad s \in \Omega \tag{2}$$

where $u(t) \in X_B[\Omega]$ is the PDE state, $c \in L_\infty^m[\Omega]$, and $\mathcal{U}$ is a vector of $m \in \mathbb{N}$ monomials in $u, u_s, \dots, u_{s_n}$

$$\mathcal{U}_i(u, u_s, \dots, u_{s_n}) := u(t,s)^{\alpha_{i0}} u_s(t,s)^{\alpha_{i1}} \cdots u_{s_n}(t,s)^{\alpha_{in}} \tag{3}$$

defined by the coefficient matrix $A := [\alpha_{ij}] \in \mathbb{N}^{m\times(n+1)}$. The PDE is assumed to have linear BCs; hence $X_B[\Omega]$ is

$$X_B[\Omega] := \left\{u \in H_2^n[\Omega] : B\begin{bmatrix}\mathcal{D}^{n-1}u(a) \\ \mathcal{D}^{n-1}u(b)\end{bmatrix} = 0, B \in \mathbb{R}^{n\times 2n}\right\} \tag{4}$$

Dirichlet, Neumann, and Robin BCs can be enforced by setting B appropriately. In summary, a polynomial PDE is parametrized by $\{c, B, A\}$ from (2)-(4).

Definition 1 (Classical solution to the PDE)**.** *For a given $u_0 \in X_B[\Omega]$, u is a classical solution to the polynomial PDE defined by $\{c, B, A\}$ if u is Frechét differentiable, $u(0) = u_0$, and (2) is satisfied by $u(t) \in X_B[\Omega]\ \forall\, t \ge 0$.*

In this section, we assume full-rank BCs (see Lem, 1) and show how a PDE of the form (2) can be defined in terms of PI operators acting on the fundamental state. The advantage of this representation is that the BCs and continuity constraints from (4) are moved into the representation of the dynamics. We now recap the definition of a PI operator [13].

Definition 2. *For a given domain $\Omega := [a,b]$ and $k \in \mathbb{N}$, the parameter space $\mathcal{N}_3^k[\Omega] := L_\infty^k[\Omega] \times L_2^k[\Omega^2] \times L_2^k[\Omega^2]$ is defined. For $x \in L_2[\Omega]$ and any $\{R_0, R_1, R_2\} \in \mathcal{N}_3^k$, the associated PI operator is defined as*

$$(\mathcal{R}x)(s) = R_0(s)x(s) + \int_a^s R_1(s,\theta)x(\theta)d\theta + \int_s^b R_2(s,\theta)x(\theta)d\theta$$

where $\mathcal{R} : L_2[\Omega] \to L_2^k[\Omega]$.

For $\{R_0, R_1, R_2\} \in \mathcal{N}_3^k$, define $\mathcal{R} := \mathcal{P}_{\{R_0,R_1,R_2\}} \in \Pi_3^k$. If $\mathcal{R} = \mathcal{P}_{\{0,R_1,R_2\}}$, then $\mathcal{R} \in \Pi_2^k$. We refer to these as 3-PI and 2-PI operators, respectively. Both Π_2^k and Π_3^k form *-algebras [23], allowing summation, composition, and adjoint operations to be computed directly from their parameters.

Now consider solutions to (2). Since $u(t) \in X_B \subset H_2^n\ \forall\, t \ge 0$, the nth order spatial derivative exists and need only satisfy $u_{s_n}(t) \in L_2$. Hence, it is free of BCs and continuity constraints. We refer to this derivative $v(t) := u_{s_n}(t)$ as the *fundamental state*. A map from the PDE state space, X_B, to the fundamental state space, L_2, is defined by the differential operator $\frac{\partial^n}{\partial s^n} : X_B \to L_2$. Assuming full-rank BCs, an inverse map exists and is defined by a 2-PI operator $\mathcal{T} : L_2 \to X_B$. We recall the following Lemma from [24] which provides explicit formulae mapping B to the operators $\mathcal{T}$, as well as $\mathcal{R}_j := \partial_s^j \circ \mathcal{T}$.

Lemma 1. *Let $\Omega := [a,b]$ and X_B be as in (4), where $B_1, B_2 \in \mathbb{R}^{n\times n}$ such that $B = [B_1\ B_2]$. For $s \in \Omega$ and $i,j \in \{0,\dots,n-1\}$, define*

$$G_{ij}(s) := \begin{cases} \frac{(s-a)^{j-i}}{(j-i)!} & j \ge i \\ 0 & else \end{cases} \qquad h_i(s) := \frac{s^{n-i-1}}{(n-i-1)!} \tag{5}$$

and set $G(s) := [G_{ij}(s)]$, $H(s) := [h_0(s)\ \dots\ h_{n-1}(s)]'$.

If $B_1 + B_2G(b)$ is full rank, then the inverse operator $\mathcal{T} : L_2[\Omega] \to X_B[\Omega]$ exists with $\mathcal{T} = \mathcal{P}_{\{T_1,T_2\}} \in \Pi_2$, where

$$\begin{aligned} T_1(s,\theta) &:= h_0(s-\theta) - G_{0,:}(s)F(\theta) \\ T_2(s,\theta) &:= -G_{0,:}(s)F(\theta) \end{aligned} \tag{6}$$

and $F(\theta) := (B_1 + B_2G(b))^{-1}B_2H(b-\theta)$. Defining $\mathcal{R}_j := \frac{\partial^j}{\partial s^j} \circ \mathcal{T} \in \Pi_2$, we have for all $u \in X_B[\Omega]$ and $v \in L_2[\Omega]$

$$u_{s_j}(s) = (\mathcal{R}_j[\frac{\partial^n}{\partial s^n}u])(s) \quad (\mathcal{R}_j v)(s) = (\frac{\partial^j}{\partial s^j}[\mathcal{T}v])(s) \tag{7}$$

Proof. Refer to Thm. 10 and Thm. 12 in [24]. □

Using this Lemma, we can express the PDE state and it's derivatives in terms of the fundamental state. Clearly, an identity operator $\mathcal{R}_n = \mathcal{P}_{\{1,0,0\}} \in \Pi_3$ can also be defined such that $u_{s_n}(s) = (\mathcal{R}_n v)(s)$. Substituting these relations into (2), we find that if u satisfies the PDE, then v satisfies

$$(\mathcal{T}v)_t(t,s) = c(s)^\top \mathcal{U}\left(\mathcal{T}v, (\mathcal{T}v)_s, \dots, (\mathcal{T}v)_{s_n}\right) \tag{8}$$

where $v(t) \in L_2[\Omega]\ \forall\, t \ge 0$. This PIE representation is free of BCs and continuity constraints, and is only represented in terms of the fundamental state. In the next section, we define the representation of a distributed polynomial and show how (8) can be expressed in that format.

IV. Polynomials on a Distributed State

In this section, we aim to generalize [15, Section IV] to enable the representation of higher order tensor products of PI operators, such as $(\mathcal{T}v)^{\otimes\alpha_{i0}}(\mathcal{R}_1 v)^{\otimes\alpha_{i1}}\dots(\mathcal{R}_n v)^{\otimes\alpha_{in}}$ from (8), and integrals over these higher order tensor products, which arise in the representation of LFs and distributed semi-algebraic sets. We first define the distributed polynomial in its most general form and then define two classes of operators to parametrize the distributed polynomial such that it can represent both terms. The key advantage of the distributed polynomial representation is that it is unique, which will later enable us to enforce equality constraints.

Definition 3. *For a given domain $\Omega := [a,b]$, we define a degree $d \in \mathbb{N}$ distributed polynomial on $x \in L_2[\Omega]$ as*

$$p(x) := \mathcal{C}\mathcal{Z}_d(x)$$

where $\mathcal{Z}_d(x)$ is the distributed monomial basis on x and $\mathcal{C} := [\mathcal{C}_1 \ \cdots \ \mathcal{C}_d]$ with $\mathcal{C}_i$ being an operator acting on the degree i distributed monomial.

A. Tensor-PI Operators

The class of Tensor-PI operators is used to represent tensor products of 3-PI operators, mapping elements of $L_2[\Omega^d]$ to elements of $L_2^k[\Omega]$.

Definition 4. *For a given domain $\Omega := [a,b]$ and $d,k \in \mathbb{N}$, the operator $\mathcal{H} : L_2[\Omega^d] \to L_2^k[\Omega]$ is a Tensor-PI operator if there exist $m, k_{ij} \in \mathbb{N}$ and 3-PI operators $\mathcal{R}_{ij} : L_2[\Omega] \to L_2^{k_{ij}}[\Omega]$ such that*

$$(\mathcal{H}[x_1 \otimes \cdots \otimes x_d])(s) = \sum_{i=1}^{m} (\mathcal{R}_{i1}x_1)(s) \otimes \cdots \otimes (\mathcal{R}_{id}x_d)(s)$$

for all $x_j \in L_2[\Omega]$, where $k = \prod_{j=1}^{d} k_{ij} \ \forall\, i$.

The operator $\mathcal{H}$ in Def. 4 is denoted by $\mathcal{H} = \sum_{i=1}^{m} \mathcal{H}_i$ where $\mathcal{H}_i = \mathcal{R}_{i1} \otimes \cdots \otimes \mathcal{R}_{id}$[1] or $\mathcal{H}_i = \mathcal{R}_{i1}^{\otimes d}$ if the operators $\mathcal{R}_{i1}, \dots, \mathcal{R}_{id}$ are the same. This includes all operators that can equally be represented as a linear combination of: (i) operators that act on a tensor product (ii) tensor products of operators. If an operator $\mathcal{H}$ satisfies Def. 4, then $\mathcal{H}$ is said to be a T-PI operator where $\mathcal{H} \in \Pi_T^k$. This representation enables tensor products of PI operators to be represented as distributed polynomials.

B. Functional-PI Operators

Functional operators map function-valued objects to real scalar values. Although such functionals $\mathcal{K} : L_2[\Omega] \to \mathbb{R}$ can always be expressed as an integral over any bounded kernel, $\mathcal{K}w := \int_\Omega K(\theta)w(\theta)d\theta$, for our purposes, the kernel is restricted to being polynomial. This class of operators is used to parametrize the distributed polynomial such that it can represent LFs and distributed semi-algebraic sets.

Definition 5. *For a given domain $\Omega := [a,b]$ and $d \in \mathbb{N}$, the operator $\mathcal{K} : L_2[\Omega^d] \to \mathbb{R}$ is a Functional-PI operator if there exist $K_\alpha \in \mathbb{R}[\boldsymbol{\theta}_d]$ for $\alpha \in A_d$ such that*

$$\mathcal{K}x = \sum_{\alpha \in A_d} \int_{\Omega^d} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) K_\alpha(\boldsymbol{\theta}_d) x(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d \quad \forall\, x \in L_2[\Omega^d] \tag{9}$$

Note that A_d and $\mathbb{I}_\alpha(\boldsymbol{\theta}_d)$ are defined in Section II. If an operator $\mathcal{K}$ satisfies Def. 5, then $\mathcal{K}$ is said to be an F-PI operator where $\mathcal{K} \in \Pi_F$. Integrals over T-PI operators are not obviously in the form of an F-PI operator; however, Lem. 2 shows they are.

Lemma 2. *For a given domain $\Omega := [a,b]$ and $d \in \mathbb{N}$, then any integrable function $K : \Omega^d \to \mathbb{R}$ satisfies*

$$\int_{\Omega^d} K(\boldsymbol{\theta}_d)\, d\boldsymbol{\theta}_d = \sum_{\alpha \in A_d} \int_{\Omega^d} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) K(\boldsymbol{\theta}_d)\, d\boldsymbol{\theta}_d \tag{10}$$

Proof. Refer to Appendix I-A for proof. □

Since integral operators are linear, it is not difficult to see that the class of F-PI operators is closed under linear combinations. In particular, if $\mathcal{K}, \mathcal{G} : L_2[\Omega^d] \to \mathbb{R}$ are F-PI operators defined by parameters $\{K_\alpha\}$ and $\{G_\alpha\}$, respectively, then $\lambda\mathcal{K} + \mu\mathcal{G} \in \Pi_F$ for any $\lambda, \mu \in \mathbb{R}$, defined by $\{\lambda K_\alpha + \mu G_\alpha\}$. In addition, a tensor product of F-PI operators can be defined as follows, acting on tensor products of distributed monomials.

Definition 6. *For a given domain $\Omega := [a,b]$ and $d,r \in \mathbb{N}$, let $\mathcal{K} : L_2[\Omega^d] \to \mathbb{R}$ and $\mathcal{G} : L_2[\Omega^r] \to \mathbb{R}$ be F-PI operators, respectively defined by parameters $\{K_\alpha\}_{\alpha \in A_d}$ and $\{G_\alpha\}_{\alpha \in A_r}$. We define $\mathcal{K} \otimes \mathcal{G} : L_2[\Omega^{d+r}] \to \mathbb{R}$ by*

$$(\mathcal{K} \otimes G)x = \sum_{\alpha \in A_{d+r}} \int_{\Omega^{d+r}} \mathbb{I}_\alpha(\boldsymbol{\theta}_{d+r}) H_{\sigma(\alpha)}(\boldsymbol{\theta}_{d+r}) x(\boldsymbol{\theta}_{d+r}) d\boldsymbol{\theta}_{d+r}$$

where $H_{\sigma(\alpha)}(\boldsymbol{\theta}_d, \boldsymbol{\theta}_r) = K_{\sigma_1(\alpha)}(\boldsymbol{\theta}_d) G_{\sigma_2(\alpha)}(\boldsymbol{\theta}_r)$ for each $\alpha \in A_d$, with $\sigma_1(\alpha) := \alpha \setminus \{d+1, \dots, d+r\} \in A_d$ and $\sigma_2(\alpha) := \alpha \setminus \{1, \dots, d\} - d \in A_r$.

Given this tensor product of F-PI operators, the following Lemma shows that the product of distributed polynomials, parametrized by such F-PI operators, can be expressed in terms of a tensor product of the F-PI operators, acting on a tensor product of the distributed monomials.

Lemma 3. *For given $\Omega := [a,b]$, $d \in \mathbb{N}$, define $\mathcal{K} := [\mathcal{K}_1 \ \cdots \ \mathcal{K}_d]$ and $\mathcal{G} := [\mathcal{G}_1 \ \cdots \ \mathcal{G}_d]$ by F-PI operators $\mathcal{K}_i, \mathcal{G}_i : L_2[\Omega^i] \to \mathbb{R}$. Let $\mathcal{C} := [0 \ \ \mathcal{C}_2 \ \cdots \ \mathcal{C}_{2d}]$ where $\mathcal{C}_k := \sum_{(i,j) \in \mathcal{S}_k} \mathcal{K}_i \otimes \mathcal{G}_j$ for $k \in \{2, \dots, 2d\}$. If $p(x) = \mathcal{K}\mathcal{Z}_d(x)$ and $q(x) = \mathcal{G}\mathcal{Z}_d(x)$, then $p(x)q(x) = \mathcal{C}\mathcal{Z}_{2d}(x) \ \forall\, x \in L_2[\Omega]$.*

Proof. Refer to Appendix I-B. □

By Lemma 3, the class of distributed polynomials parametrized by F-PI operators is closed under the multiplication operation, thus forming a ring (like polynomials in finite variables). We now define the subset of F-PI operators with the particular form which arises from integrals over T-PI operators. This is referred to as the "vectorization" of scalar T-PI operators.

[1] Formally, this notation is more akin to the face-splitting product.

Definition 7. *For $\Omega := [a,b]$, $i \in \{1,\dots,d\}$ and a given set of polynomial parameters $R_{i,1}, R_{i,2} \in \mathbb{R}[s,\theta_i]$, define the associated functions $H_\alpha \in \mathbb{R}[\boldsymbol{\theta}_d]$ for each $\alpha \in A_d$ as*

$$H_\alpha(\boldsymbol{\theta}_d) := \sum_{j=0}^{d} \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} R_{1,\gamma(1,j)}(s,\theta_1)\cdots R_{d,\gamma(d,j)}(s,\theta_d)\,ds$$

where $\theta_{\alpha_0} := a$, $\theta_{\alpha_{d+1}} := b$. For a given tensor product $\mathcal{H} := \mathcal{R}_1 \otimes \cdots \otimes \mathcal{R}_d : L_2[\Omega^d] \to L_2[\Omega]$, where each $\mathcal{R}_i = \mathcal{P}_{\{R_{i,1},R_{i,2}\}} \in \Pi_2$, define $\mathrm{vec}(\mathcal{H}) : L_2[\Omega^d] \to \mathbb{R}$ as

$$\mathrm{vec}(\mathcal{H})x := \sum_{\alpha\in A_d} \int_{\Omega^d} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) H_\alpha(\boldsymbol{\theta}_d) x(\boldsymbol{\theta}_d)\,d\boldsymbol{\theta}_d$$

Then, for a given scalar T-PI operator $\mathcal{H} := \sum_{i=1}^m \mathcal{H}_i$ where $\mathcal{H}_i = \mathcal{R}_{i,1} \otimes \cdots \otimes \mathcal{R}_{i,d} : L_2[\Omega^d] \to L_2[\Omega]$, we define $\mathrm{vec}(\mathcal{H}) := \sum_{j=1}^m \mathrm{vec}(\mathcal{H}_i)$.

Note that the indicator functions $\gamma(\alpha_k, j)$, $\mathbb{I}_\alpha(\boldsymbol{\theta}_d)$ are defined in Section II. Finally, the following Lemma proves that the vectorization of a scalar T-PI operator corresponds to the integral over the scalar T-PI operator, enabling integrals over scalar T-PI operators to be represented as distributed polynomials.

Lemma 4. *For a given domain $\Omega := [a,b]$, any scalar T-PI operator $\mathcal{H} : L_2[\Omega^d] \to L_2[\Omega]$ satisfies*

$$\int_\Omega (\mathcal{H}x)(s)\,ds = \mathrm{vec}(\mathcal{H})x \qquad \forall\, x \in L_2[\Omega^d]$$

Proof. Refer to Appendix I-C for proof. □

Linear combinations and multiplication of distributed polynomials, parametrized by F-PI operators, arise when constructing the distributed SOS program for local stability analysis of polynomial PDEs. The closure and vectorization results presented in this section enable the equality constraints to be enforced.

C. Example: Distributed Semi-algebraic Sets

In the upcoming sections, we show how T-PI and F-PI operators can be used to parametrize distributed polynomials for the representation of polynomial PDEs and LFs. For now, we present an example of how these operators can be used to represent distributed semi-algebraic sets, such as the L_2-ball of radius r, as distributed polynomials.

For a domain $\Omega := [a,b]$ and $r > 0$, define the ball of radius r in the PDE domain

$$X_{B,r}[\Omega] := \left\{u \in X_B[\Omega] :\ r^2 - \|u\|_{L_2}^2 \geq 0\right\} \tag{11}$$

Using the inverse map $\mathcal{T} : L_2[\Omega] \to X_B[\Omega] \in \Pi_2$, applying Def. 4, and Lemma 4, the constraint is equal to

$$\begin{aligned} r^2 - \|\mathcal{T}v\|_{L_2}^2 &= r^2 - \int_\Omega (\mathcal{T}v)(s)(\mathcal{T}v)(s)ds \\ &= r^2 - \int_\Omega \left((\mathcal{T}\otimes\mathcal{T})v^{\otimes 2}\right)(s)ds \\ &= r^2 - \underbrace{\mathrm{vec}\,(\mathcal{T}\otimes\mathcal{T})}_{\in\Pi_F} v^{\otimes 2} = \mathcal{G}\mathcal{Z}_2(v) \end{aligned}$$

a distributed polynomial in the fundamental state, where $\mathcal{G} = [r^2 \ \ 0 \ \ -\mathrm{vec}(\mathcal{T}\otimes\mathcal{T})]$. As a result, the set (11) can be represented by the following distributed semi-algebraic set on the fundamental state

$$L_{2,r}[\Omega] := \{v \in L_2[\Omega] :\ \mathcal{G}\mathcal{Z}_2(v) \geq 0\} \tag{12}$$

V. A PIE Representation of Polynomial PDEs

We now revisit the PIE representation of the evolution of v in (8). Using the T-PI operators introduced in the previous section, this PIE can be expressed as a distributed polynomial. In particular, we show that the evolution of v is governed by a distributed polynomial of the form

$$\mathcal{T}v_t(t,s) = \mathcal{C}\mathcal{Z}_d(v) \quad s \in \Omega \tag{13}$$

where $v(t) \in L_2[\Omega]$, $\mathcal{T} \in \Pi_2$, and $\mathcal{C} := [\mathcal{C}_1 \ \dots \ \mathcal{C}_d]$ is parametrized by T-PI operators $\mathcal{C}_k : L_2[\Omega^k] \to L_2[\Omega] \in \Pi_T$. With reference to the polynomial PDE defined by (2)-(4), the operators $\mathcal{C}_k$ are given by

$$\mathcal{C}_k := \sum_{i=1}^{m} \mathbf{1}(k-\alpha_i)\mathcal{M}_{c_i} \circ \mathcal{H}_i \tag{14}$$

where $\alpha_i := \sum_{j=0}^n \alpha_{ij}$ denotes the sum over row i of matrix A; $d = \max\{\alpha_1,\dots,\alpha_m\}$; $\mathcal{M}_{c_i} := \mathcal{P}_{\{c_i,0,0\}}$ is the multiplier operator associated to $c_i \in L_\infty[\Omega]$; and $\mathcal{H}_i := \mathcal{T}^{\otimes\alpha_{i0}} \otimes \mathcal{R}_1^{\otimes\alpha_{i1}} \otimes \cdots \otimes \mathcal{R}_n^{\otimes\alpha_{in}}$ is the T-PI operator associated to the monomial $\mathcal{U}_i$, also defined by A. We note here $\mathcal{M}_{c_i} \circ \mathcal{H}_i \in \Pi_T$ follows from the fact that Π_T is closed under linear combinations.

Definition 8 (Classical solution to the PIE)**.** *For a given $v_0 \in L_2[\Omega]$, v is a classical solution to the polynomial PIE defined by $\{\mathcal{T},\mathcal{C}\}$ if v is Frechét differentiable, $v(0) = v_0$, and* (13) *is satisfied by $v(t) \in L_2[\Omega]\ \forall\, t \geq 0$.*

The following Lemma proves there exists an invertible map between classical solutions to the polynomial PDE (2) and classical solutions to the associated PIE (13).

Lemma 5. *Suppose that $B \in \mathbb{R}^{n\times 2n}$ satisfies Lemma 1 and for $j \in \{0,\dots,n\}$ the operators $\mathcal{T},\mathcal{R}_j \in \Pi_3$ are defined accordingly. Let $c \in L_\infty^m[\Omega]$, $A \in \mathbb{N}^{m\times(n+1)}$ and define the operators $\mathcal{C}_k$ as in* (14)*. Then, v is a classical solution to the PIE, defined by $\{\mathcal{T},\mathcal{C}\}$, with initial state v_0 if and only if $\mathcal{T}v$ is a classical solution to the PDE, defined by $\{c,B,A\}$, with initial state $\mathcal{T}v_0$. Conversely, u is a classical solution to the polynomial PDE, defined by $\{c,B,A\}$, with initial state u_0 if and only if u_{s_n} is a classical solution to the polynomial PIE, defined by $\{\mathcal{T},\mathcal{C}\}$, with initial state $(u_0)_{s_n}$.*

Proof. By Lemma 1, the operators $\mathcal{T},\mathcal{R}_j \in \Pi_3$ for $j \in \{0,1,\dots,n\}$ exist and are defined, along with operators $\mathcal{C}_k \in \Pi_T$ for $k \in \{1,\dots,d\}$, as defined in (14). Exploiting these definitions and that of a T-PI operator, the PIE becomes

$$\begin{aligned} (\mathcal{T}v)_t(t,s) &= \sum_{k=1}^{d} (\mathcal{C}_k[v^{\otimes k}])(t,s) = \sum_{i=1}^{m} (\mathcal{M}_{c_i} \circ \mathcal{H}_i[v^{\otimes\alpha_i}])(\cdot) \\ &= \sum_{i=1}^{m} c_i(s)(\mathcal{T}v)^{\alpha_{i0}}(\cdot)(\mathcal{R}_1 v)^{\alpha_{i1}}(\cdot)\dots(\mathcal{R}_n v)^{\alpha_{in}}(\cdot) \end{aligned}$$

$$\mathcal{Q} := \begin{bmatrix} \mathcal{Q}_{11} & \mathcal{Q}_{12} & \cdots & \mathcal{Q}_{1d} \\ \mathcal{Q}_{21} & \mathcal{Q}_{22} & \cdots & \mathcal{Q}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{Q}_{d1} & \mathcal{Q}_{d2} & \cdots & \mathcal{Q}_{dd} \end{bmatrix} := \begin{bmatrix} \hat{\mathcal{U}} & & & \\ & \hat{\mathcal{U}}^{\otimes 2} & & \\ & & \ddots & \\ & & & \hat{\mathcal{U}}^{\otimes d} \end{bmatrix}^* \circ \underbrace{\begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \cdots & \bar{Q}_{1d} \\ \bar{Q}_{21} & \bar{Q}_{22} & \cdots & \bar{Q}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{Q}_{d1} & \bar{Q}_{d2} & \cdots & \bar{Q}_{dd} \end{bmatrix}}_{:=\bar{Q}} \circ \underbrace{\begin{bmatrix} \hat{\mathcal{U}} & & & \\ & \hat{\mathcal{U}}^{\otimes 2} & & \\ & & \ddots & \\ & & & \hat{\mathcal{U}}^{\otimes d} \end{bmatrix}}_{:=\hat{U}_d} \tag{16}$$

Also by Lemma 1, relations (7) are leveraged to express the PIE in the same form as the PDE.

$$(\mathcal{T}v)_t(t,s) = c(s)^\top \mathcal{U}\left(\mathcal{T}v, (\mathcal{T}v)_s, \ldots, (\mathcal{T}v)_{s_n}\right)$$

It follows that, for any $v_0 \in L_2[\Omega]$, v is a classical solution to the PIE defined by $\{\mathcal{T}, \mathcal{C}\}$ if and only if $\mathcal{T}v_0 \in X_B[\Omega]$ and Tv is a classical solution to the PDE defined by $\{c, B, A\}$.

Conversely, for $j \in \{0, 1, \ldots, n\}$ the operators $\mathcal{T}, \mathcal{R}_j \in \Pi_3$ exist and are defined by Lemma 1. Using the relations (7), the PDE becomes

$$\begin{aligned}(\mathcal{T}u_{s_n})_t(t,s) &= c(s)^\top \mathcal{U}(\mathcal{T}u_{s_n}, \mathcal{R}_1 u_{s_n}, \ldots, \mathcal{R}_n u_{s_n}) \\ &= \sum_{i=1}^{m} c_i(s)(\mathcal{T}u_{s_n})^{\alpha_{i0}}(\mathcal{R}_1 u_{s_n})^{\alpha_{i1}} \cdots (\mathcal{R}_n u_{s_n})^{\alpha_{in}}\end{aligned}$$

Defining operators $\mathcal{H}_i$, $\mathcal{M}_i$, and $\mathcal{C}_k$, as in (14), results in the PDE (2) expressed in the same form as the PIE (13).

$$(\mathcal{T}u_{s_n})_t(t,s) = \sum_{i=1}^{m} (\mathcal{M}_{c_i} \circ \mathcal{H}_i[u_{s_n}^{\otimes \alpha_i}])(t,s) = \mathcal{C}\mathcal{Z}_d(u_{s_n})$$

It follows that for a given $u_0 \in X_B[\Omega]$, u is a classical solution to the PDE, defined by $\{c, B, A\}$, if and only if $(u_0)_{s_n} \in L_2[\Omega]$ and u_{s_n} is a classical solution to the PIE, defined by $\{\mathcal{T}, \mathcal{C}\}$. □

VI. SOS Polynomials on a Distributed State

Inspired by SOSTOOLS [22], which builds on SOS polynomials for the analysis of polynomial ODEs, we introduce the notion of a distributed SOS polynomial. This class allows us to enforce positivity of distributed polynomials by definition and through equality constraints. In the subsequent section, we employ distributed SOS polynomials as LFs and formulate an SOS program to certify exponential stability of the equilibrium solution, $u \equiv 0$, on an L_2-ball of radius r.

Definition 9. *For a given domain $\Omega := [a, b]$, let the vector of degree $\bar{d} \in \mathbb{N}$ monomials in $s, \theta \in \Omega$ be denoted by $\mathcal{U}_{\bar{d}}(s,\theta) \in \mathbb{R}^{\mu(\bar{d})}[s,\theta]$, where $\mu(\bar{d}) := \frac{1}{2}(\bar{d}+1)(\bar{d}+2)$. Define*

$$(\hat{\mathcal{U}}x)(s) := \begin{bmatrix} \int_a^s \mathcal{U}_{\bar{d}}(s,\theta)x(\theta)\,d\theta \\ \int_s^b \mathcal{U}_{\bar{d}}(s,\theta)x(\theta)\,d\theta \end{bmatrix} \in \Pi_2^{2\mu(\bar{d})}$$

We say that $q : L_2[\Omega] \to \mathbb{R}$ is a distributed SOS polynomial in x, of degree $2d \in \mathbb{N}$, if there exists $\bar{d} \in \mathbb{N}$ and $\bar{Q} \succeq 0$ of appropriate dimension such that

$$q(x) := \left\langle \mathcal{Z}_d(x), \mathcal{Q}\mathcal{Z}_d(x) \right\rangle_{L_2} \tag{15}$$

with $\mathcal{Q}$ defined in (16)*. In which case, we write $q \in \Sigma_{2d}$.*

It should be observed that $\hat{\mathcal{U}}^{\otimes i} : L_2[\Omega^i] \to L_2^{(2\mu(\bar{d}))^i}[\Omega]$ is a T-PI operator; as a result, each matrix $\bar{Q}_{ij}$ has dimension $(2\mu(\bar{d}))^i \times (2\mu(\bar{d}))^j$. Furthermore, the distributed SOS polynomial can be expressed as a sum of quadratic forms

$$q(x) = \sum_{i,j=1}^{d} \left\langle x^{\otimes i}, \mathcal{Q}_{ij} x^{\otimes j} \right\rangle_{L_2} \quad \mathcal{Q}_{ij} : L_2[\Omega^j] \to L_2[\Omega^i]$$

By parametrizing distributed SOS polynomials in a quadratic manner, as in Def. 9, it is clear that any $q \in \Sigma_{2d}$ satisfies $q(x) \geq 0$ for all $x \in L_2[\Omega]$. However, since the quadratic representation of polynomials is not unique, it must be converted to the linear form for enforcing equality constraints. The remainder of this section shows how to perform this conversion, thereby also verifying that elements of Σ_{2d} are indeed distributed polynomials as per Def. 3. The next definition shows how the individual quadratic terms can be evaluated by defining the "vectorization" of a distributed SOS polynomial.

Definition 10. *For a given domain $\Omega := [a, b]$ and operator $\mathcal{Q} : L_2[\Omega^j] \to L_2[\Omega^i]$ of the form*

$$\mathcal{Q} = \underbrace{(\hat{\mathcal{U}}_{L,1} \otimes \cdots \otimes \hat{\mathcal{U}}_{L,i})^*}_{:=\hat{\mathcal{U}}_L} \circ \bar{Q} \circ \underbrace{(\hat{\mathcal{U}}_{R,1} \otimes \cdots \otimes \hat{\mathcal{U}}_{R,j})}_{:=\hat{\mathcal{U}}_R}$$

where $\hat{\mathcal{U}}_{L,\ell} \in \Pi_2^m$, $\hat{\mathcal{U}}_{R,r} \in \Pi_2^n$, and $\bar{Q} \in \mathbb{R}^{m^i \times n^j}$. Define

$$\text{vec}(\mathcal{Q}) := \sum_{p=1}^{m^i} \sum_{q=1}^{n^j} \bar{Q}_{pq} \text{vec}\left((\hat{\mathcal{U}}_L)_p \otimes (\hat{\mathcal{U}}_R)_q\right)$$

It should be observed that $\hat{\mathcal{U}}_L, \hat{\mathcal{U}}_R \in \Pi_T$. Combining Def. 10 and Lemma 4, each quadratic term can be converted to the standard form of a distributed polynomial (Def. 3). This is summarised in the following Lemma.

Lemma 6. *For a given domain $\Omega := [a, b]$ and any $\mathcal{Q} : L_2[\Omega^j] \to L_2[\Omega^i]$ as defined in Def. 10*

$$\langle z, \mathcal{Q}y \rangle_{L_2} = \text{vec}(\mathcal{Q})(z \otimes y) \tag{17}$$

$\forall\, z \in L_2[\Omega^i]$ and $y \in L_2[\Omega^j]$.

Proof. By Def. 10 and expanding the inner product, for any $z \in L_2[\Omega^i]$ and $y \in L_2[\Omega^j]$

$$\begin{aligned}\langle z, \mathcal{Q}y \rangle_{L_2} &= \left\langle \hat{\mathcal{U}}_L z, \bar{Q} \circ \hat{\mathcal{U}}_R y \right\rangle_{L_2} \\ &= \sum_{p=1}^{m^i} \sum_{q=1}^{n^j} \bar{Q}_{pq} \int_\Omega ((\hat{\mathcal{U}}_L)_p z)(s)((\hat{\mathcal{U}}_R)_q y)(s) ds\end{aligned}$$

As $(\hat{\mathcal{U}}_L)_p : L_2[\Omega^i] \to L_2[\Omega]$ and $(\hat{\mathcal{U}}_R)_q : L_2[\Omega^j] \to L_2[\Omega]$ are both scalar T-PI operators, applying Def. 4 leads to

$$= \sum_{p=1}^{m^i} \sum_{q=1}^{n^j} \bar{Q}_{pq} \int_\Omega \left(\left((\hat{\mathcal{U}}_L)_p \otimes (\hat{\mathcal{U}}_R)_q\right)[z \otimes y] \right)(s) ds$$

Applying Lemma 4

$$= \sum_{p=1}^{m^i} \sum_{q=1}^{n^j} \bar{Q}_{pq} \text{vec}\big((\hat{\mathcal{U}}_L)_p \otimes (\hat{\mathcal{U}}_R)_q\big)[z \otimes y]$$

followed by Def. 10 results in (17). □

This section concludes by converting the distributed SOS polynomial, defined in quadratic form, to the standard (linear) form of Def. 3.

Lemma 7. *For a given domain $\Omega := [a, b]$, a distributed SOS polynomial $q \in \Sigma_{2d}$, as defined in Def. 9, satisfies*

$$q(x) = \text{vec}(\mathcal{Q})\mathcal{Z}_{2d}(x) \tag{18}$$

where $\text{vec}(\mathcal{Q}) := [0 \;\; \mathcal{Q}_2 \;\; \cdots \;\; \mathcal{Q}_{2d}]$ contains F-PI operators $\mathcal{Q}_w : L_2[\Omega^w] \to \mathbb{R}$ of the form

$$\mathcal{Q}_w = \sum_{(i,j)\in\mathcal{S}_w} \text{vec}(\mathcal{Q}_{ij})$$

where the F-PI operators, $\text{vec}(\mathcal{Q}_{ij})$, are such that $\left\langle x^{\otimes i}, \mathcal{Q}_{ij} x^{\otimes j} \right\rangle_{L_2} = \text{vec}(\mathcal{Q}_{ij})(x^{\otimes i} \otimes x^{\otimes j}) \;\; \forall\, x \in L_2[\Omega]$.

Proof. Starting from Def. 9 and applying Lemma 6, with $\text{vec}(\mathcal{Q}_{ij})$ defined in Def. 10,

$$q(x) = \sum_{i,j=1}^{d} \text{vec}(\mathcal{Q}_{ij})(x^{\otimes i} \otimes x^{\otimes j})$$

Since F-PI operators are closed under linear combinations, operators acting on the same degree distributed monomial can be summed (i.e, summing over $\mathcal{S}_w$ as defined in Section II) to get (18). □

Armed with Lemma 7, we can now define LFs as distributed SOS polynomials and enforce negativity of its derivative through an equality constraint, as shown next.

VII. Distributed SOS Program For Local Stability Analysis of Polynomial PDEs

In the previous sections, we have shown how polynomial PIEs and semi-algebraic sets can be represented as distributed polynomials. Furthermore, we have also parameterized a class of nonnegative distributed polynomials, as distributed SOS polynomials. Using these results, we now show how local stability of the equilibrium $u \equiv 0$ for a PDE as in (2) can be verified in the associated PIE representation (13), by solving a distributed SOS program. Specifically, to verify stability, we look for a distributed SOS polynomial LF, $V(v)$, which decays towards zero on a given semi-algebraic region. We test this decay by verifying negativity of the Lie derivative, defined as follows.

Definition 11. *For a given polynomial PIE defined by $\{\mathcal{T}, \mathcal{C}\}$ and distributed polynomial, $V(v)$, define the Lie derivative $\mathcal{L}_{\mathcal{T},\mathcal{C}}V(v)$ such that $\mathcal{L}_{\mathcal{T},\mathcal{C}}V(v(t)) = \frac{d}{dt}V(v(t))$ for any classical solution $v(t)$ to the PIE.*

Choosing $V \in \Sigma_{2d}$, the Lie derivative $\mathcal{L}_{\mathcal{T},\mathcal{C}}V(v)$ along the polynomial PIE in (13) will be a distributed polynomial as well. In order to evaluate this derivative, we assume V to be of the form $V(v) = \hat{V}(\mathcal{T}v) = \hat{V}(u)$, where $\hat{V}(u)$ is then a LF for the PDE. For simplicity, the following Lemma provides an explicit expression for the Lie derivative of a degree 2 distributed SOS LF. Of course, this can also be derived for arbitrary degree $d \in \mathbb{N}$, but the proof is significantly more cumbersome.

Lemma 8. *For a given polynomial PIE defined by $\{\mathcal{T}, \mathcal{C}\}$, let $\mathcal{C} := \begin{bmatrix}\mathcal{C}_1 & \cdots & \mathcal{C}_{d'}\end{bmatrix}$. For a self-adjoint 3-PI operator $\mathcal{P} = \mathcal{P}^*$, let $V(v) := \langle \mathcal{T}v, \mathcal{P}\mathcal{T}v\rangle_{L_2}$. Then*

$$\mathcal{L}_{\mathcal{T},\mathcal{C}}V(v) = 2\sum_{i=1}^{d'} \text{vec}\left(\mathcal{C}_i \otimes (\mathcal{P} \circ \mathcal{T})\right)[v^{\otimes i+1}]$$

Proof. Let $v(t)$ be an arbitrary classical solution to the PIE defined by $\{\mathcal{T}, \mathcal{C}\}$. Then, by Lemma 4,

$$\begin{aligned}
\frac{d}{dt}V(v(t)) &= 2\left\langle \mathcal{T}v_t(t), \mathcal{P}\mathcal{T}v(t)\right\rangle_{L_2} \\
&= 2\left\langle \mathcal{C}\mathcal{Z}_d(v(t)), \mathcal{P}\mathcal{T}v(t)\right\rangle_{L_2} \\
&= 2\sum_{i=1}^{d'} \int_\Omega (\mathcal{C}_i v(t,\cdot)^{\otimes i})(s)(\mathcal{P}\mathcal{T}v(t,\cdot))(s)\, ds \\
&= 2\sum_{i=1}^{d'} \int_\Omega \left(\mathcal{C}_i \otimes (\mathcal{P}\circ\mathcal{T})\right)[v(t,\cdot)^{\otimes i+1}](s)\, ds \\
&= 2\sum_{i=1}^{d'} \text{vec}(\mathcal{C}_i \otimes (\mathcal{P}\circ\mathcal{T}))v(t)^{\otimes i+1}
\end{aligned}$$

□

Given that the Lie derivative of a distributed polynomial is again a distributed polynomial, negativity of this derivative can be tested by searching for a distributed SOS polynomial, $W \in \Sigma_{2d}$, such that $\mathcal{L}_{\mathcal{T},\mathcal{C}}V = -W$. In this manner, existence of a LF certifying stability on the L_2-ball of radius r can be formulated as a distributed SOS program, yielding the following Theorem.

Theorem 1. *For a given polynomial PDE with domain X_B as in (4) satisfying the conditions of Lem. 1, let $\{\mathcal{T}, \mathcal{C}\}$ denote the associated polynomial PIE. For $r > 0$, let $g_r(v) := r^2 - \text{vec}(\mathcal{T}\otimes\mathcal{T})v^{\otimes 2}$. For some $\epsilon, C > 0$, $\lambda \geq 0$, and $d, d' \in \mathbb{N}$, where $d = d' + 1$, suppose there exists $V \in \Sigma_{2d}$, $p_1, p_2 \in \Sigma_{2d'}$, such that*

$$\begin{aligned}
V(v) - \epsilon^2 \text{vec}(\mathcal{T}\otimes\mathcal{T})v^{\otimes 2} &\in \Sigma_{2d} \\
C^2\text{vec}(\mathcal{T}\otimes\mathcal{T})v^{\otimes 2} - V(v) - p_1(v)g_r(v) &\in \Sigma_{2d} \\
-\mathcal{L}_{\mathcal{T},\mathcal{C}}V(v) - 2\lambda V(v) - p_2(v)g_r(v) &\in \Sigma_{2d}
\end{aligned}$$

Then, for $M := C/\epsilon$ and any $u_0 \in X_{B,r/M}$, as defined in (11), solutions to the PDE with initial state $u(0) = u_0$ satisfy

$$\|u(t)\|_{L_2} \leq Me^{-\lambda t}\|u_0\|_{L_2} \;\; \forall\, t \geq 0$$

Proof. Suppose there exists V satisfying the proposed conditions, and let $\hat{V}(u) := V(u_{s_n})$ for $u \in X_B$, so that $\hat{V}(\mathcal{T}v) = V(v)$. An $\epsilon > 0$ must exist such that $\hat{V}(u) \geq \epsilon\|u\|_{L_2}^2$; hence, a sufficiently small $\gamma > 0$ can always be chosen such that $\hat{V}(u) \leq \gamma$, which implies $\|u\|_{L_2}^2 \leq r^2$. That is, the γ-level set of $\hat{V}(u)$ is contained in $X_{B,r}$. Now

TABLE I

LARGEST LOWER BOUND ON DECAY RATE λ ENSURING EXPONENTIAL STABILITY OF THE FISHER EQ. (19) ON AN L_2-BALL OF RADIUS r.

r	0.01	0.05	0.1	0.25	0.5	1	2	3	4
λ_r	4.857	4.810	4.734	4.572	4.270	3.650	2.692	1.388	0.047

define $L_{2,r}[\Omega] := \{v \in L_2[\Omega] : \ g_r(v) \geq 0\}$ and suppose the distributed SOS program is feasible. Then, for all $v \in L_{2,r}$

$$V(v) \geq \epsilon^2 \text{vec}(\mathcal{T} \otimes \mathcal{T}) v^{\otimes 2} \geq \epsilon^2 \|\mathcal{T} v\|_{L_2}^2$$
$$V(v) \leq C^2 \text{vec}(\mathcal{T} \otimes \mathcal{T}) v^{\otimes 2} - p_1(v) g(v) \leq C^2 \|\mathcal{T} v\|_{L_2}^2$$
$$\mathcal{L}_{\mathcal{T},\mathcal{C}} V(v) \leq -2\lambda V(v) - p_2(v) g(v) \leq 0$$

For $\gamma > 0$, define the γ^2-sublevel set of V as

$$S_\gamma(V) := \{v \in L_2[\Omega] : \ V(v) \leq \gamma^2\}$$

Since $V(v) \geq \epsilon^2 \|\mathcal{T} v\|_{L_2}^2$ for all $v \in L_2[\Omega]$, it follows that for all $v \in S_{\epsilon r}$

$$\|\mathcal{T} v\|_{L_2}^2 \leq \frac{1}{\epsilon^2} V(v) \leq r^2$$

whence $S_{\epsilon r} \subseteq L_{2,r}$. Since $\dot{V}(v) \leq 0$ for all $v \in L_{2,r}$, this implies that for any solution $v(t)$ to the PIE, $v(0) \in S_{\epsilon r}$ implies $v(t) \in L_{2,r}$ for all $t \geq 0$. Furthermore, since $V(v) \leq C^2 \|\mathcal{T} v\|_{L_2}^2$, we have for any $v_0 \in L_{2,r/M}$ that

$$V(v_0) \leq C^2 \|\mathcal{T} v\|_{L_2}^2 \leq r^2 \epsilon^2$$

whence $L_{2,r/M} \subseteq S_{\epsilon r}$. It follows that, if $v(0) = v_0 \in L_{2,r/M}$, then $v(t) \in S_{\epsilon r} \subseteq L_{2,r}$ and therefore $\mathcal{L}_{\mathcal{T},\mathcal{C}} V(v(t)) \leq -2\lambda V(v(t))$ for all $t \geq 0$. By the Grönwall-Bellman inequality, $v_0 \in L_{2,r/M}$ implies

$$V(v(t)) \leq e^{-2\lambda t} V(v_0) \quad \forall\, t \geq 0$$

Now, for any $u_0 \in X_{B,r/M}$, let $u(t) \in X_B$ be the associated solution to the PDE for $t \geq 0$. Then $u(t) = \mathcal{T} v(t)$ for all $t \geq 0$, where $v(t) := u_{s_n}(t) \in L_2[\Omega]$ is a solution to the PIE, with $v_0 := (u_0)_{s_n}$. Since $u_0 \in X_{B,r/M}$, we have $\|u_0\|_{L_2} = \|\mathcal{T} v_0\|_{L_2} \leq r/M$, and therefore $v_0 \in L_{2,r/M}$. It follows that, for all $t \geq 0$

$$\begin{aligned}\|u(t)\|_{L_2} = \|\mathcal{T} v(t))\|_{L_2} &\leq \frac{1}{\epsilon} \sqrt{V(v(t))} \\ &\leq \frac{1}{\epsilon} e^{-\lambda t} \sqrt{V(v_0)} \\ &\leq \frac{C}{\epsilon} e^{-\lambda t} \|\mathcal{T} v_0\|_{L_2} = M e^{-\lambda t} \|u_0\|_{L_2}\end{aligned}$$

□

VIII. NUMERICAL EXAMPLE

In this section, we apply the distributed SOS program (Thm. 1) to the Fisher Equation with Dirichlet BCs.

$$\begin{aligned} u_t(t,s) &= u_{ss}(t,s) + \alpha u(t,s) - \beta u(t,s)^2 \\ u(t,0) &= u(t,1) = 0 \quad t \geq 0 \ \ s \in [0,1] \end{aligned} \tag{19}$$

We consider this system with parameters $(\alpha, \beta) = (5, -1)$, where the origin of the linearization is exponentially stable with rate $\lambda = 4.869...$. However, for the nonlinear system, the term $-u(t,s)^2$ dominates when $\|u(t)\|_{L_2}$ becomes too large, so the origin is only locally stable. To verify this numerically, we construct the PIE representation as

$$\mathcal{T} v_t(t,s) = (\mathcal{C}_1 v)(t,s) + (\mathcal{C}_2[v^{\otimes 2}])(t,s)$$

where $v := u_{ss}$, $\mathcal{C}_1 = 1 + 5\mathcal{T} \in \Pi_3$, $C_2 = -\mathcal{T}^{\otimes 2}$, and

$$(\mathcal{T} v)(s) := \int_0^s (s-1)\theta v(\theta)\, d\theta + \int_s^1 s(\theta - 1) v(\theta)\, d\theta$$

We can verify local stability of the Fisher Equation by solving the distributed SOS program in Thm. 1. Specifically, we fix $\lambda = 0$ and declare a quadratic LF, $V(v) = \langle \mathcal{T} v, \mathcal{Q} \mathcal{T} v \rangle_{L_2} \in \Sigma_2$, where $\mathcal{Q} = \hat{\mathcal{Q}} + \epsilon^2 I$ for $\hat{Q}$ as in Def 9 with $d = 1$ and $\bar{d} = 4$. Performing bisection on r, feasibility of the distributed SOS program (Thm. 1) can then be verified up to $r = 4.048$. In particular, feasibility can be verified with the energy functional, $V(v) = \langle \mathcal{T} v, \mathcal{T} v \rangle_{L_2}$, with $M = 1$, certifying stability on the L_2-ball of radius $r = 4.048$. Simulating the PDE with initial condition $u_0(s) = r\sqrt{2}\sin(\pi s)$ (so that $\|u_0\|_{L_2} = r$), instability is observed for $r > 4.05$, suggesting the obtained lower bound $r = 4.048$ is tight.

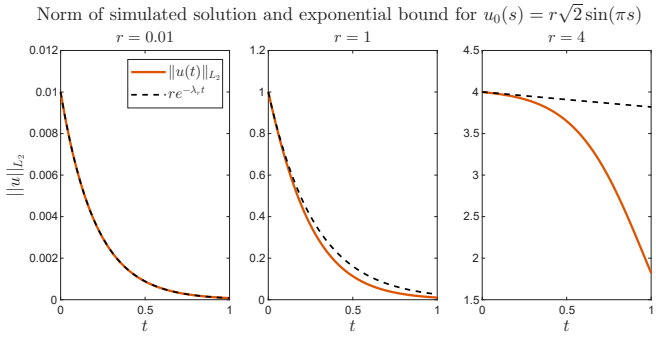


Fig. 1. Norm of simulated solutions to the Fisher Equation (19) for $u_0(s) := r\sqrt{2}\sin(\pi s)$, along with an exponential upper bound $re^{-\lambda_r t}$ with decay rates λ_r as in Table I.

Thm. 1 is also applied to verify exponential stability of the PDE (19) on the L_2-ball of varying radius r, in the PDE domain. The largest lower bound, λ, on the exponential decay rate is established by performing bisection on λ. The results are given in Table I. For each value of r and λ, the obtained value of M in Thm. 1 is roughly equal to 1, certifying that $\|u_0\|_{L_2} \leq r$ implies $u(t) \leq e^{-\lambda t} \|u_0\|_{L_2}$ for all $t \geq 0$. The simulated solutions for $u_0(s) = r\sqrt{2}\sin(\pi s)$ with $r \in \{0.8, 2.4, 4.0\}$ are displayed in Fig. 1, along with the associated exponential bounds $e^{-\lambda t}$. The results show that the obtained λ provide a tight lower bound on the initial decay rate of the solution starting with $\|u_0\|_{L_2} \leq r$. As the value of r decreases to 0, the lower bound λ also increases to the rate $\lambda = 4.869...$ of the linearized system, as expected.

IX. CONCLUSION

In this paper, we proposed the distributed polynomial representation. This compact and flexible representation can model the PIE, associated with the polynomial PDE, when parametrized by Tensor-PI operators. When parametrized by Functional-PI operators, it can represent distributed semi-algebraic sets and distributed SOS LFs. To enforce positivity of the distributed SOS LF and negativity of its derivative, a distributed SOS polynomial was also defined. Leveraging

these representations, a distributed SOS program was constructed for testing local stability over the L_2-ball of radius r, as demonstrated on the Fisher Equation. Whilst currently limited to scalar and 1D PDEs, the results of this paper may be extended to coupled PDEs in higher dimensions.

APPENDIX I
PROOFS

A. Proof of Lemma 2

Proof. The result follows by induction. For the base case $d = 1$, the result holds trivially since $\mathbb{I}_1(\theta_1) = 1$. Now, suppose the relation holds for some $d \in \mathbb{N}$. Then, for $d+1$

$$\begin{aligned}
&\int_{\Omega^{d+1}} K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} d\boldsymbol{\theta}_d = \int_{\Omega^d}\int_{\Omega} K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} d\boldsymbol{\theta}_d \\
&= \sum_{\alpha\in A_d} \int_{\Omega^d}\int_{\Omega} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} d\boldsymbol{\theta}_d \\
&= \sum_{\alpha\in A_d} \int_{\Omega^d} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) \left[\int_a^{\theta_{\alpha_1}} K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} \right. \\
&\qquad + \sum_{j=1}^{d-1} \int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}} K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} \\
&\qquad \left. + \int_{\theta_{\alpha_d}}^{b} K(\theta_{d+1}, \boldsymbol{\theta}_d)\, d\theta_{d+1} \right] d\boldsymbol{\theta}_d \\
&= \sum_{\alpha'\in A_{d+1}} \int_{\Omega^{d+1}} \mathbb{I}_{\alpha'}(\boldsymbol{\theta}_{d+1}) K(\boldsymbol{\theta}_{d+1})\, d\boldsymbol{\theta}_{d+1}
\end{aligned}$$

□

B. Proof of Lemma 3

Proof. Fix arbitrary $i, j \in \{1, \dots, d\}$, and let $\mathcal{K}_i : L_2[\Omega^i] \to \mathbb{R}$ and $\mathcal{G}_j : L_2[\Omega^j] \to \mathbb{R}$ be defined by parameters $\{K_\alpha\}_{\alpha\in A_i}$ and $\{G_\beta\}_{\beta\in A_j}$, respectively. Then, for all $x \in L_2[\Omega]$,

$$\begin{aligned}
&\left(\mathcal{K}_i x^{\otimes i}\right)\left(\mathcal{G}_j x^{\otimes j}\right) \\
&= \sum_{\alpha\in A_i}\sum_{\beta\in A_j} \int_{\Omega^i}\int_{\Omega^j} \mathbb{I}_\alpha(\boldsymbol{\theta}_i)\mathbb{I}_\beta(\boldsymbol{\eta}_j) H_{\alpha,\beta}(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j) w(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j)\, d\boldsymbol{\eta}_j d\boldsymbol{\theta}_i
\end{aligned}$$

where we define $H_{\alpha,\beta}(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j) := K_\alpha(\boldsymbol{\theta}_i) G_\beta(\boldsymbol{\eta}_j)$ and $w \in L_2[\Omega^{i+j}]$. Now, for each $\alpha \in A_i$ and $\beta \in A_j$, define Ξ as the set of all unions $\alpha \cup (\beta + i)$ that preserve the ordering of α and β,

$$\Xi_{\alpha,\beta} := \{\gamma \in A_{i+j} \mid \sigma_1(\gamma) = \alpha,\ \sigma_2(\gamma) - i = \beta\}, \quad (20)$$

where $\sigma_1(\gamma) := \gamma \setminus \{i+1, \dots, i+j\} \in A_i$ and $\sigma_2(\gamma) := \gamma \setminus \{1, \dots, i\} - i \in A_j$. Then, for any $\gamma \in A_{i+j}$,

$$\mathbb{I}_\gamma(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j)\mathbb{I}_\alpha(\boldsymbol{\theta}_i)\mathbb{I}_\beta(\boldsymbol{\eta}_j) = \begin{cases} \mathbb{I}_\gamma(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j) & \text{if } \gamma \in \Xi_{\alpha,\beta} \\ 0 & \text{else} \end{cases}$$

and therefore, by Lem. 2,

$$\begin{aligned}
&\int_{\Omega^i}\int_{\Omega^j} \mathbb{I}_\alpha(\boldsymbol{\theta}_i)\mathbb{I}_\beta(\boldsymbol{\eta}_j) H_{\alpha,\beta}(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j) w(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j)\, d\boldsymbol{\eta}_j d\boldsymbol{\theta}_i \\
&= \sum_{\gamma\in A_{i+j}} \int_{\Omega^i}\int_{\Omega^j} \mathbb{I}_\gamma(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j)\mathbb{I}_\alpha(\boldsymbol{\theta}_i)\mathbb{I}_\beta(\boldsymbol{\eta}_j) \\
&\qquad\qquad H_{\alpha,\beta}(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j) w(\boldsymbol{\theta}_i, \boldsymbol{\eta}_j)\, d\boldsymbol{\eta}_j d\boldsymbol{\theta}_i \\
&= \sum_{\gamma\in\Xi_{\alpha,\beta}} \int_{\Omega^{i+j}} \mathbb{I}_\gamma(\boldsymbol{\theta}_{i+j}) H_{\sigma_1(\gamma),\sigma_2(\gamma)}(\boldsymbol{\theta}_{i+j}) w(\boldsymbol{\theta}_{i+j})\, d\boldsymbol{\theta}_{i+j}
\end{aligned}$$

Noting that $\bigcup_{\alpha\in A_i}\bigcup_{\beta\in A_j} \Xi_{\alpha,\beta} = A_{i+j}$, this implies

$$\begin{aligned}
&\left(\mathcal{K}_i x^{\otimes i}\right)\left(\mathcal{G}_j x^{\otimes j}\right) = \sum_{\alpha\in A_i}\sum_{\beta\in A_j}\sum_{\gamma\in\Xi_{\alpha,\beta}} \int_{\Omega^{i+j}} \mathbb{I}_\gamma(\boldsymbol{\theta}_{i+j}) \\
&\qquad\qquad H_{\sigma_1(\gamma),\sigma_2(\gamma)}(\boldsymbol{\theta}_{i+j}) x^{\otimes i+j}(\boldsymbol{\theta}_{i+j})\, d\boldsymbol{\theta}_{i+j} \\
&= \sum_{\gamma\in A_{i+j}} \int_{\Omega^{i+j}} \mathbb{I}_\gamma(\boldsymbol{\theta}_{i+j}) H_{\sigma_1(\gamma),\sigma_2(\gamma)}(\boldsymbol{\theta}_{i+j}) x^{\otimes i+j}(\boldsymbol{\theta}_{i+j})\, d\boldsymbol{\theta}_{i+j} \\
&= (\mathcal{K}_i \otimes \mathcal{G}_j) x^{\otimes i+j}.
\end{aligned}$$

Now, let $p(x) = \mathcal{K}\mathcal{Z}_d(x)$ and $q(x) = \mathcal{G}\mathcal{Z}_d(x)$. Then, it follows that

$$\begin{aligned}
p(x)q(x) = \mathcal{K}\mathcal{Z}_d(x)\, \mathcal{G}\mathcal{Z}_d(x) &= \sum_{i=1}^{d}\sum_{j=1}^{d} (\mathcal{K}_i x^{\otimes i})(\mathcal{G}_j x^{\otimes j}) \\
&= \sum_{i=1}^{d}\sum_{j=1}^{d} (\mathcal{K}_i \otimes \mathcal{G}_j) x^{\otimes i+j} = \mathcal{C}\mathcal{Z}_{2d}(x).
\end{aligned}$$

where $\mathcal{C} = [0\ \ \mathcal{C}_2\ \ \cdots\ \ \mathcal{C}_{2d}]$, $\mathcal{C}_k = \sum_{(i,j)\in\mathcal{S}_k} \mathcal{K}_i \otimes \mathcal{G}_j$, and $\mathcal{S}_k$ is defined in Section II. □

C. Proof of Lemma 4

Proof. Let $\Omega := [a, b]$. For $i \in \{1, \dots, d\}$ and a given set of polynomial parameters $R_{i,1}, R_{i,2} \in \mathbb{R}[s, \theta_i]$, define the set of scalar 2-PI operators $\mathcal{R}_i = \mathcal{P}_{\{R_{i,1}, R_{i,2}\}} \in \Pi_2$. Now, let $\mathcal{H} = \mathcal{R}_1 \otimes \cdots \otimes \mathcal{R}_d : L_2[\Omega^d] \to L_2[\Omega]$. Then,

$$\begin{aligned}
&\int_\Omega (\mathcal{H}x)(s) ds = \int_\Omega (\mathcal{R}_1 x_1)(s) \cdots (\mathcal{R}_d x_d)(s)\, ds \\
&= \sum_{\beta\in\{1,2\}^d} \int_\Omega\int_{\Omega^d} \prod_{i=1}^{d} \left[\mathbf{I}^{\beta_i}(s, \theta_i) R_{i,\beta_i}(s, \theta_i) x_i(\theta_i) d\theta_i\right] ds
\end{aligned}$$

where $\mathbf{I}^1(s, \theta_i) = \mathbf{I}_{0^+}(s - \theta_i)$, $\mathbf{I}^2(s, \theta_i) = \mathbf{I}_{0^+}(\theta_i - s)$. Defining $x(\boldsymbol{\theta}_d) := x_d(\theta_d) \cdots x_1(\theta_1)$, and applying Lemma 2, where $K(\boldsymbol{\theta}_d) = M(\boldsymbol{\theta}_d) x(\boldsymbol{\theta}_d)$, it follows that

$$\begin{aligned}
&= \int_{\Omega^d} \overbrace{\left[\sum_{\beta\in\{1,2\}^d} \int_\Omega \prod_{i=1}^{d} \mathbf{I}^{\beta_i}(s, \theta_i) R_{i,\beta_i}(s, \theta_i) ds\right]}^{:= M(\boldsymbol{\theta}_d)} x(\boldsymbol{\theta}_d)\, d\boldsymbol{\theta}_d \\
&= \sum_{\alpha\in A_d} \int_{\Omega^d} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) M(\boldsymbol{\theta}_d) x(\boldsymbol{\theta}_d) d\boldsymbol{\theta}_d
\end{aligned}$$

Now, consider an arbitrary $\alpha \in A_d$. Then, we can decompose

$$\begin{aligned}
&\mathbb{I}_\alpha(\boldsymbol{\theta}_d) M(\boldsymbol{\theta}_d) \\
&= \sum_{\beta\in\{1,2\}^d} \sum_{j=0}^{d} \int_{\theta_j}^{\theta_{j+1}} \mathbb{I}_\alpha(\boldsymbol{\theta}_d) \prod_{i=1}^{d} [\mathbf{I}^{\beta_i}(s, \theta_i) R_{i,\beta_i}(s, \theta_i)] ds
\end{aligned}$$

where $\theta_0 := a$ and $\theta_{d+1} := b$. By definition of $\gamma(\alpha_k, j)$ (Section II) and $\mathbf{I}^{\beta_i}(s, \theta_i)$, it follows that

$$\mathbb{I}_\alpha(\boldsymbol{\theta}_d)M(\boldsymbol{\theta}_d)=\sum_{j=0}^{d}\int_{\theta_{\alpha_j}}^{\theta_{\alpha_{j+1}}}\prod_{i=1}^{d}R_{i,\gamma(i,j)}(s,\theta_i)\,ds := Q_\alpha(\boldsymbol{\theta}_d)$$

Subbing $Q_\alpha(\boldsymbol{\theta}_d)$ back into the full expression results in

$$\begin{aligned}\int_\Omega(\mathcal{H}x)(s)ds &= \sum_{\alpha\in A_d}\int_{\Omega^d}Q_\alpha(\boldsymbol{\theta}_d)x(\boldsymbol{\theta}_d)d\boldsymbol{\theta}_d\\ &= \mathrm{vec}(\mathcal{H})x \qquad \forall\, x\in L_2[\Omega^d]\end{aligned}$$

□

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